

Day 6

JavaScript notes

JavaScript is a programming language used to make web pages interactive. In the browser, it can respond to clicks, validate forms, update content, and work with APIs. JavaScript also runs outside the browser through Node.js, which lets you build server-side apps and use packages from npm.

Key basics:

- Variables: `let`, `const`, and `var`.
- Data types: string, number, boolean, null, undefined, object, array.
- Common features: functions, loops, conditionals, events, objects, arrays.
- In the browser, JavaScript works with the DOM to read and change page elements.

Node.js notes

Node.js is a JavaScript runtime that lets JavaScript run outside the browser. It is commonly used for backend development, command-line tools, automation scripts, and package management with npm. The Node.js distribution includes npm, and the official docs recommend using a version manager like `nvm` when possible.

Useful commands:

- `node -v` to check Node.js version.
- `npm -v` to check npm version.
- `node app.js` to run a JavaScript file in Node.js.

Installation steps

1. Go to the official Node.js download page and install the LTS version for your OS.
2. On Windows, you can use the installer or `nvm-windows` if you want easier version management.
3. After installation, open a new terminal so `node` and `npm` are added to your PATH.
4. Verify with `node -v` and `npm -v`. If both show version numbers, setup is correct.

Example:

```
node -v
npm -v
```

If the commands fail, the usual causes are a bad PATH setup, not reopening the terminal, or an incomplete installation.

Inspect tool notes

The Inspect tool is part of browser Developer Tools and is used to examine HTML, CSS, and JavaScript behind a webpage. You can open it by right-clicking a page and choosing **Inspect**, or with shortcuts like `Ctrl+Shift+C` / `Cmd+Shift+C` for element selection mode.

What it helps you do:

- View the DOM structure of the page.
- See and edit CSS styles live.
- Hover over items to highlight page regions.
- Click an element to jump to its HTML in the Elements panel.

Developer tools uses

Developer Tools are useful for debugging layout issues, testing JavaScript, checking network requests, and experimenting with changes without affecting the real website. In Node.js debugging, Chrome DevTools can also be opened through `chrome://inspect` for inspecting running Node processes.

Common tabs you should know:

- **Elements:** HTML and CSS.
- **Console:** JavaScript logs and errors.
- **Sources:** Debugging and breakpoints.
- **Network:** API and file requests.
- **Application:** Storage, cookies, and cache.